Choke Packets

- Used for congestion control (both VC & datagram nets)
- Router monitors utilization of output lines
- If a threshold is passed, set a warning state for the line
- Arriving packets that are routed to an output line in a warning state generate “choke packets” back on the input line they arrive on with destination specified
- Forwarded packet is tagged to prevent subsequent routers from generating a choke packet

Choke Packets (Cont.)

- Source host receiving choke packet must reduce rate of traffic sent to the specified destination
- Since packets in transit may generate several choke packets, a host can ignore choke packets for a fixed time interval after receiving the first one
- After the interval if there is still a congestion problem, the host well get more choke packets and must then reduce its rate further

Choke Packets (Cont.)

- Since this technique is feedback driven, it doesn't slow the flow when there is no congestion
- Variations include multiple warning levels and different forms of utilization (# buffers used, queue length) as trigger
- PROBLEM: what if offending host ignores the choke packets?

Hop-by-Hop Choke Packets

- Choke packet takes too long to get back to source in large WAN with high-speed ➔ source reacts slowly
- This algorithm has the choke packet affect each hop (usually a router) along the path
- The goal is to address congestion quickly at the point of greatest need ➔ propagate the “relief” back to the source

Hop-by-hop Choke Packets (Cont.)

- This generates greater need for buffers at the router
  - Required to reduce output
  - Meanwhile the input continues full blast until the choke packet propagates to the next hop

Load Shedding

- The “big hammer” - router just starts throwing out packets
- Packet discard policy may depend on the application
  - “wine” ➔ drop new packets (old wine is better than new wine) - good for file transfer
  - “milk” ➔ drop old packets (don’t even need to talk about old milk!) - good for video/multimedia

Load Shedding (Cont.)

- Requires application to mark packet with priority
  - How to keep every packet from being marked - DO NOT DISCARD ?
- Back to carrier/customer environment - make it cheaper to send LOW PRIORITY packets

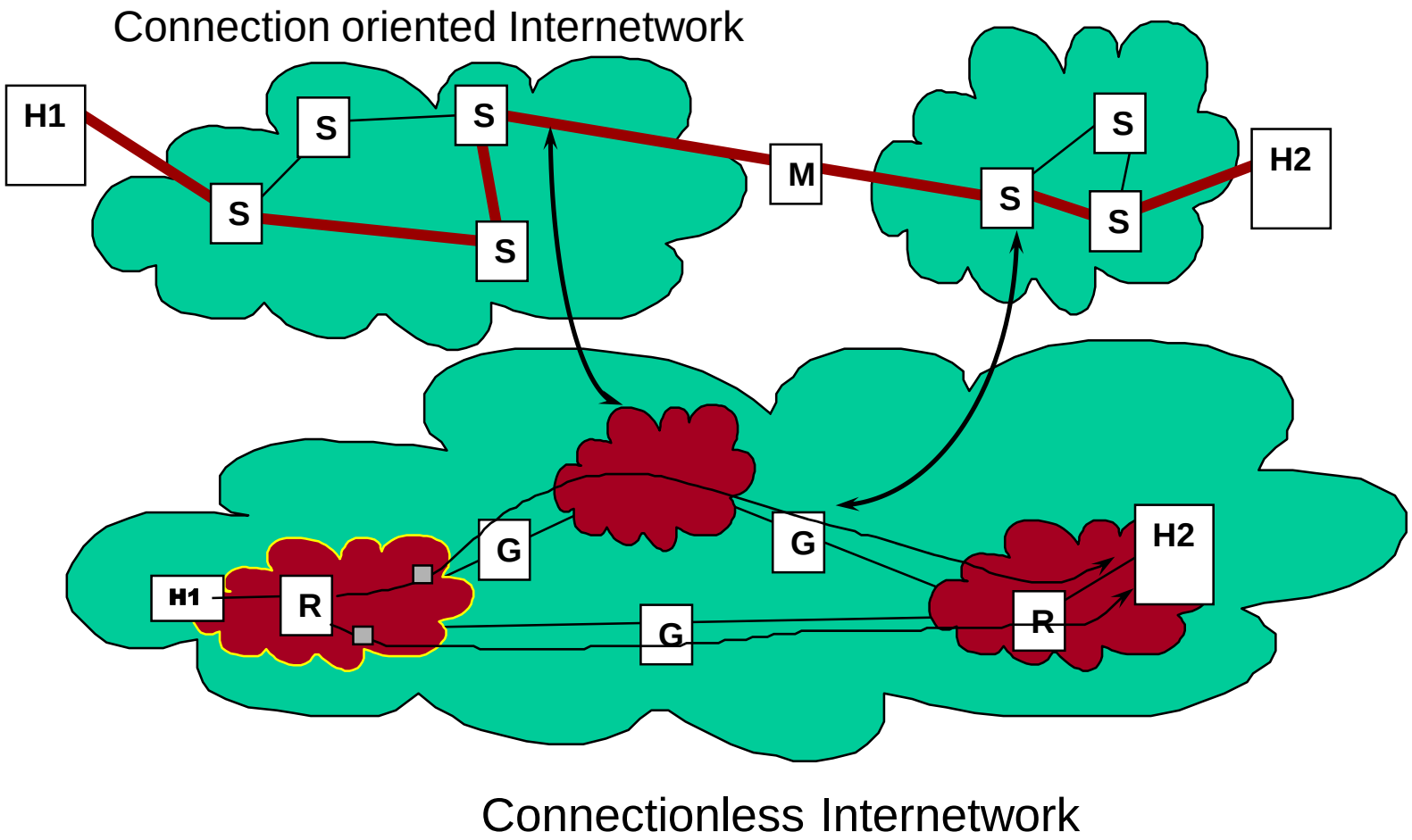
Load Shedding (Cont.)

- If service is negotiated mark, any packets that exceeded the negotiated service as low priority
- For most networks a packet may be just a portion of a message (48 byte payload of ATM cell is usually a piece of a PDU) and dropping a cell will usually result in retransmission of whole PDU
  - Drop all the cells making up that PDU
- Use early packet discard to try to preempt congestion

Internetworking Issues

- Expect that there will continue to be a large variety of protocols at each layer
- Interconnecting heterogeneous networks will introduce many conflicts
- To provide services we want the network layer to accommodate:
  - Different addressing schemes
  - Different maximum packet sizes
  - Different network access mechanisms

Internetworking Approaches



Internetworking Issues

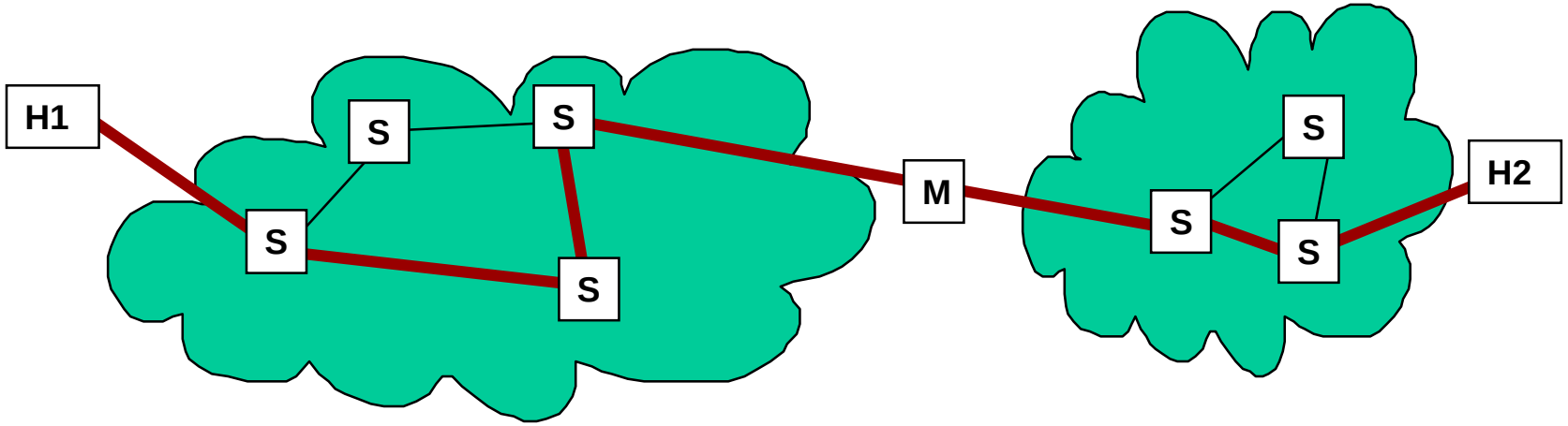
- Network layer may have to accommodate:
  - Different timeout values
  - Error recovery
  - Status reporting
  - Routing & congestion control
  - User access control
  - Service philosophy

Internetworking Approaches

- Same two competing approaches:
  - Connection oriented with virtual circuits
  - Connectionless with Datagrams

Connection Oriented Approach

- We build a virtual circuit pathway through the internetwork between the source and the destination
- Switches maintain information about VCs



Connection Oriented Approach (Cont.)

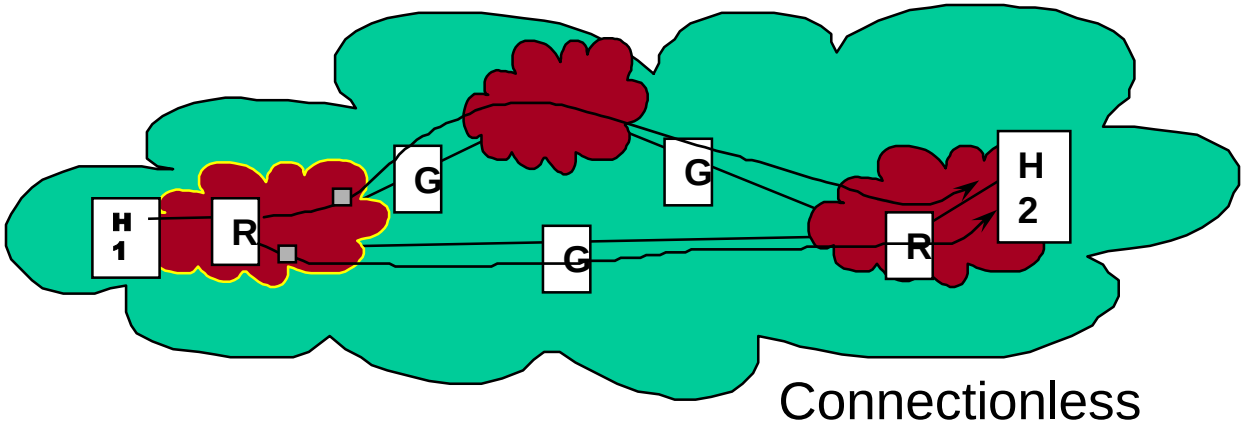
- The connection-oriented approach is often more appropriate when the internetwork is homogeneous
- Benefits of VC based internetworking:
  - Resource allocation at circuit setup
  - Sequencing is guaranteed
  - Low header overhead
  - No duplicate packets

Connection Oriented Approach (Cont.)

- Drawbacks:
  - Switch resources needed for each circuit
  - Switch failure brings down the whole connection
  - Certain paths may be susceptible to congestion
  - Difficult to incorporate non-VC based network into the internetwork

Connectionless Approach

- For connectionless we route the packets through the network with routers performing a role similar to the switches but packets do not need to all follow the same route
  - Useful for heterogeneous networks



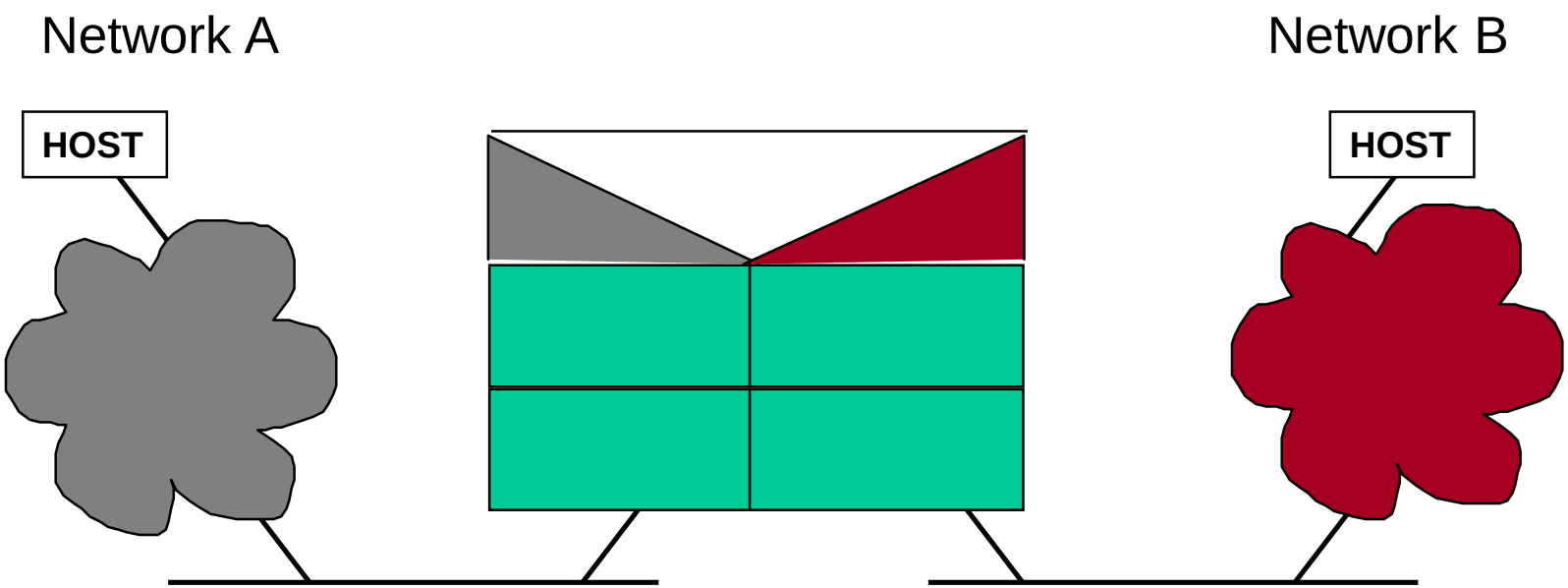
Connectionless Internetwork

Gateways

- Gateways interconnect networks with different naming/addressing conventions, depending on layer:
  - Repeaters - physical layer
  - Bridges - DL/MAC layer
  - Routers (gateways, Multiprotocol routers) - network layer
  - Transport gateways - transport layer
  - Application gateways (e.g. email gateway) - application layer

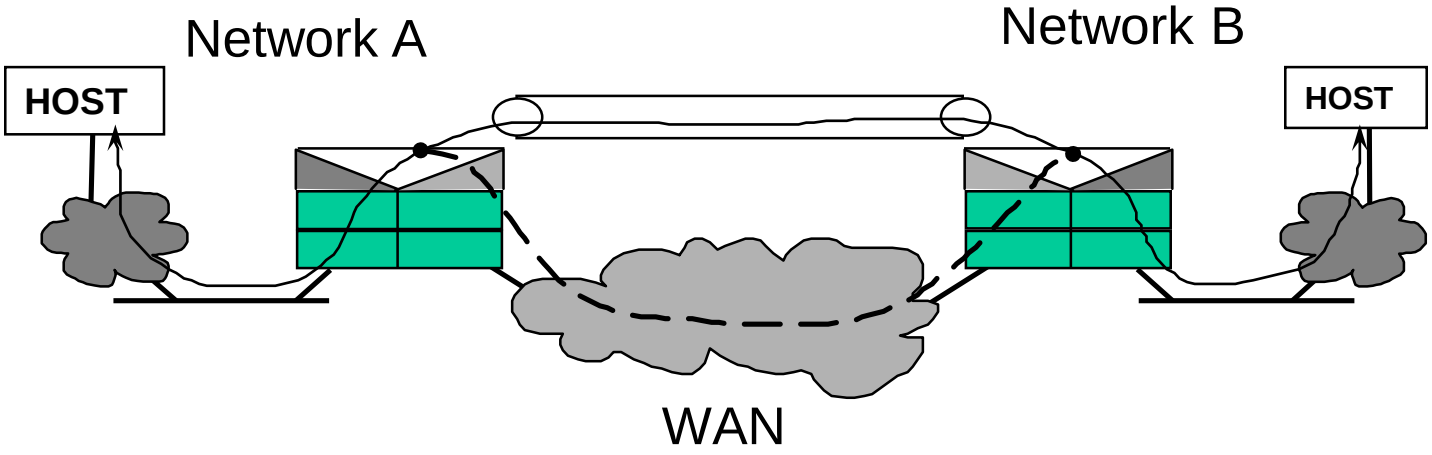
Example: Network Gateways

- Here, the gateway performs routing and translation functions between Network A and Network B



Tunneling

- Gateway does not translate to the WAN protocol between Network A and Network B but wraps the IP packet in a WAN packet and sends it transparently (tunnels) across the WAN. A & B seem to have a direct serial link.



Fragmentation

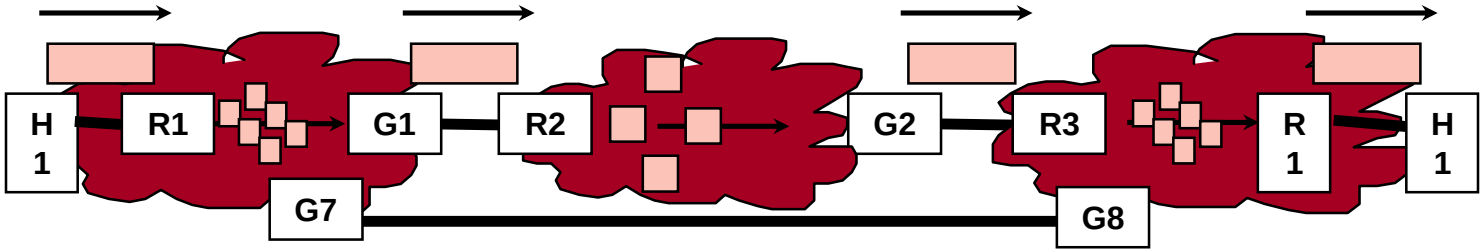
- If data has to traverse many diverse networks, it is likely that they will have different maximum data “payload” sizes
- This may be determined by the operating system parameters, protocol specifications, etc.
- Usually the size of PDU payload increases in higher layers (higher levels of abstraction)
- Internetwork has to deal with differences - usually means we have to fragment larger packets

Fragmentation (Cont.)

- Easy part - Gateway is allowed to break up a packet into fragments and send fragments separately
- Hard part - Gateway has to put pieces back together to reconstruct the original packet
- So the obvious question is - do we need to put them back together again?
- As usual there are two competing viewpoints
  - Transparent Fragmentation
  - Non-Transparent Fragmentation

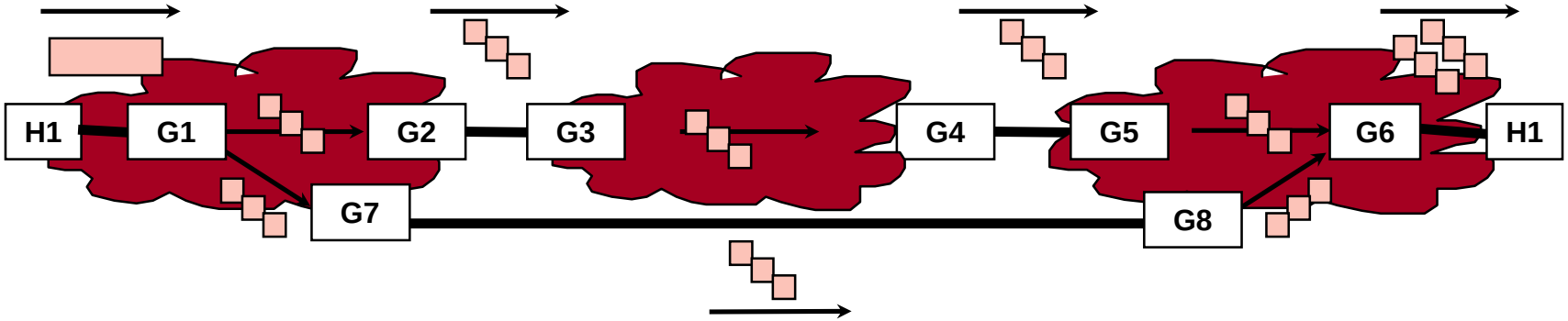
Transparent Fragmentation

- Fragments recombined at each gateway and original sized packet delivered at destination
- Requires all packets to leave network via same gateway ➔ some performance loss
- Gateway needs to know when all fragments are received



Non-transparent Fragmentation

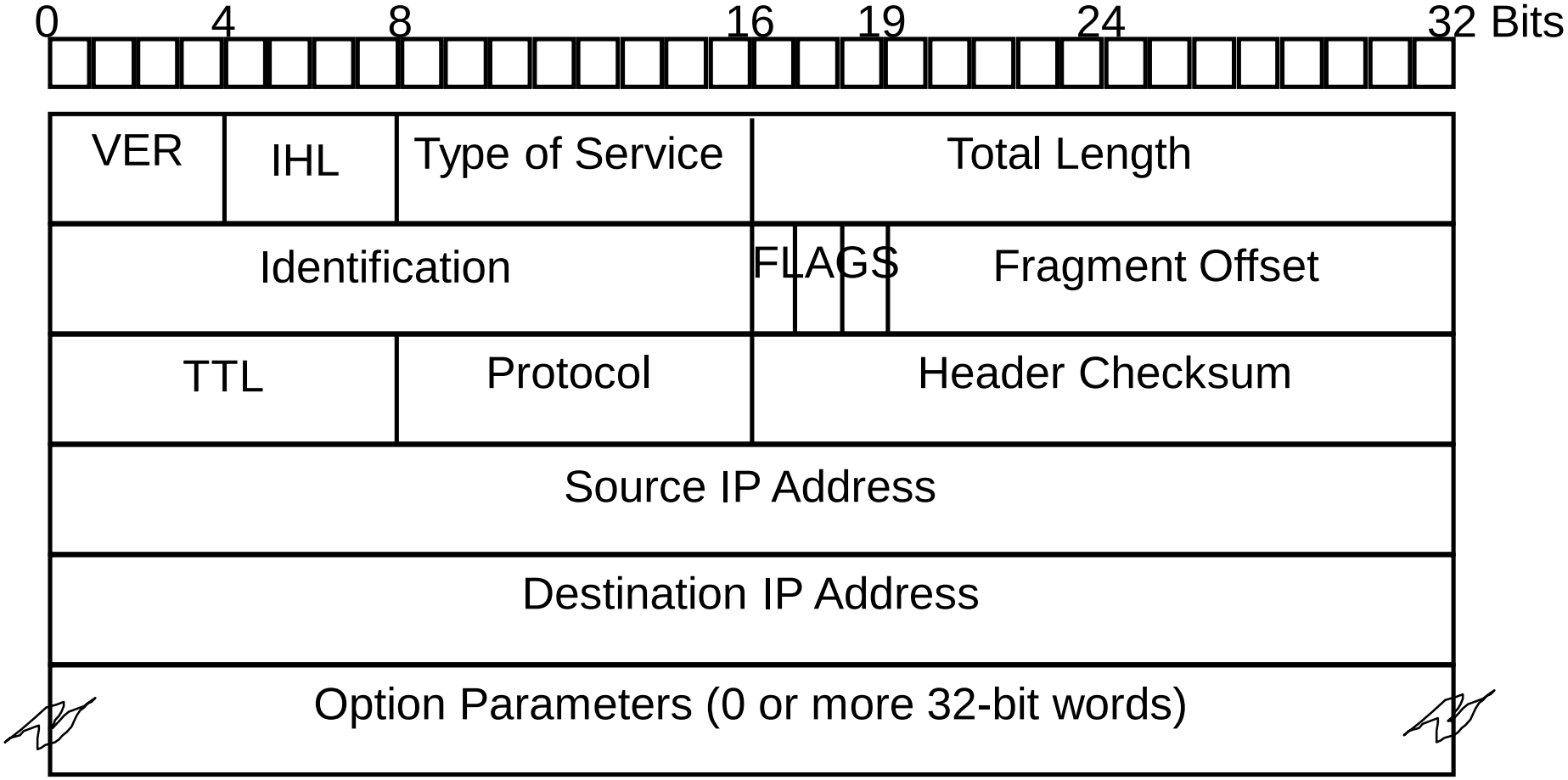
- Do not recombine fragments at each intermediate gateway ➔ each fragment becomes an independent packet
- Allows fragments to take separate paths
- Recombination takes place at the destination host



The Internet Protocol (IP)

- A collection of Autonomous Systems interconnected by one or more backbones
- Loose, collaborative structure with Autonomous Systems (AS's) organized into Regional Networks interconnected into the larger Internet
- Developed from DARPANET ➔ NSFNET ➔ Internet
- Provides best effort datagram service to Transport Layer

IP Packet Header Format



Basic IP Services

- Send and Receive services
- Send (Src Addr, Dst Addr, Protocol, Service Type, Identifier, Don’t Fragment, TTL, Len, Options, Data)
  - Src Addr = IP Address of sender
  - Dst Addr = IP Address of destination
  - Protocol = Recipient Protocol using IP Services
  - Service Type = Indicates type of service requested
  - Identifier = Combined with three above to uniquely identify data unit ...

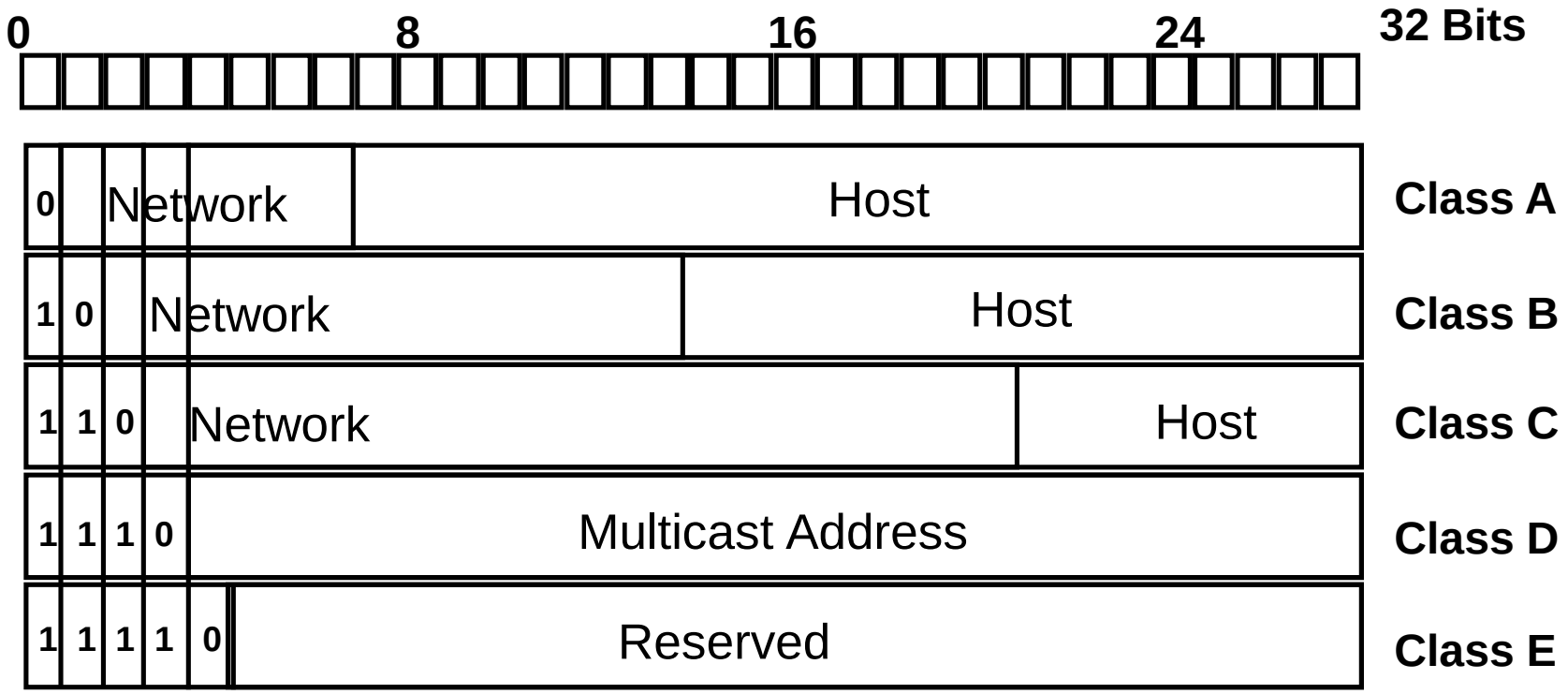
Basic IP Services – Send (Cont.)

- Don’t Fragment = Says whether or not IP is allowed to fragment data
- TTL = Time To Live for packet
- Len = Length of data being sent
- Options = Options requested by IP user
- Data = The IP user data

Options

- Allow rarely used parameters and future extensions
  - Security
  - Source routing (specify list of routers for packet)
  - Route recording (keep a record of routers visited by packet)
  - Stream ID
  - Timestamping (source and intermediate routers timestamp packet)

IP Addressing



IP Addressing - Special Addresses

0 0 0 0 0 0 …….. 0 0 0	This host
0 0 0 0 ……. 0 0 | Host	Host on this Network
1 1 1 1 1 1 1 …….. 1 1 1	Broadcast on this Network
NETWORK | 1 1 1 1 1 . . . . . 1 1 1	Broadcast on remote Network
127 | DON’T CARE	Loopback

Internet Control Message Protocol (ICMP)

- IP standards specify that compliant implementations must also implement ICMP (RFC 792)
- ICMP provides a mechanism to provide feedback about problems in the network
- ICMP packets can be sent by routers and hosts
- ICMP exists at the NL but is a user of NL services, i.e., it uses IP datagram service
- ICMP packets are usually generated by a host or router in response to a previous datagram

ICMP (Cont.)

- ICMP packets have a 64-bit header which includes:
  - Type (8 bits) - type of ICMP packet
  - Code (8 bits) - specifies parameters of the packet
  - Checksum (16 bits) - checksum for entire ICMP packet
  - Parameters (32 bits) - parameters too large for Code
- Header is usually followed by additional information depending on packet type
- When the packet refers to a previous datagram the additional info includes the IP header and first 64 bits of the original datagram

Types of ICMP Packets

- Inclusion of first 64 bits of data after the IP header is to allow IP entity to determine which IP user was associated with the datagram
- Types of packets include:
  - Destination unreachable (e.g., router can’t reach destination network)
  - Time exceeded - TTL of datagram reached zero
  - Parameter error - semantic error in IP header
  - Source quench - simple flow control

Types of ICMP Packets (Cont.)

- Redirect - advise host of a better route
- Echo (reply) - test communications
- Timestamp (reply) - allow determination of delay
- Address mask req (reply) - inform host of LAN's subnet mask

Some ICMP Packet Formats

Type	Code	Checksum
Unused		

Dst. unreachable, time exceeded, src quench

Type	Code	Checksum
Ptr	Unused	
IP Header + 64 bits original dg		

Parameter error

Type	Code	Checksum
Identifier		Sequence #
Originate timestamp		

Timestamp

Type	Code	Checksum
Identifier		Sequence #
Originate timestamp		
Receive timestamp		
Transmit timestamp		

Timestamp reply

Some ICMP Packet Formats (Cont.)

Type	Code	Checksum
Gateway IP Address		
IP Header + 64 bits original dg		

Redirect

Type	Code	Checksum
Identifier	Sequence #

Address mask request

Type	Code	Checksum
Identifier	Sequence #	
IP Header + 64 bits original dg		

Echo, Echo Reply

Type	Code	Checksum
Identifier	Sequence #	
Address Mask		

Address mask reply

Mapping IP to DL Addresses

- Consider IP layer running on an IEEE 802.3 LAN
- Recall DL has its own 48-bit addresses
- NL superimposes an internetwork on top of the LAN and provides its own 32-bit IP address space
- DL knows nothing about IP addresses
- How do these two sets of addresses get mapped to each other?

Address Resolution Protocol (ARP)

- Another control protocol which resides at the NL
- ARP builds a DL broadcast frame with a packet "what's the DL address for IP address w.x.y.z?" and sends it
- Broadcast frame is received by all hosts and one says "that's me!" or another says "I know"

Address Resolution Protocol (ARP)

- Host recognizing the IP address builds a response giving the DL address to IP address mapping and sends it to the sender of the broadcast
- Address mappings are cached to prevent repeated broadcasts
- DL-to-IP mapping of sender may be cached by all hosts on the LAN for future use

Address Resolution Protocol (ARP)

- Host may broadcast ARP for its own address upon booting as a way of announcing its mapping
- This is a simple and effective protocol which eliminates need for maintaining static tables
- Since LAN broadcasts are not routed, the router DL generally becomes the mapping for remote networks